IEEE Research Papers (2021-2026)

S.No	Year	Authors	Dataset	Modality	Methodology/ Model Used	Accuracy (%)	Gaps Identified / Limitations
1	2023	S. A. Bakry et al.	Multiple (Review)	Video, EEG	Machine Learning pipelines (CAD systems)	N/A	Lack of large-scale multimodal datasets; need for real-time clinical validation in everyday settings.
2	2023	S. T. Noor et al.	PhysioNet / Pervasive EEG	EEG, ECG	LSTM Autoencoder with RNN	97	High computational cost of LSTM frameworks; limited to specific demographics.
3	2022	Not Specified	Not Specified	EEG	Sentiment Analysis via Brain Signals	Not Specified	Vulnerability to noise and motion artifacts in EEG signals; small subject pools.
4	2012	S. Koelstra et al.	DEAP Database	EEG, Physiological, Video	Single-trial classification & Affective state estimation	Baseline	Subjective rating bias (self-reporting); laboratory stimuli don't perfectly reflect natural environments.
5	2024	M. M. R. Khan et al.	Math Task (23 Subjects)	EEG	SVM and Ensemble Learning with optimized pre-processing	96.85	Handcrafted features limit automated scalability; tested exclusively on mathematical stressors.
6	2023	S. K. Yadav et al.	Audio/ Speech Samples	Audio, EEG	Filter Feature Selection (RELIEFF), KNN, Naive Bayes	88	Data synchronization issues between audio and EEG modalities; relatively lower accuracy.
7	2025	Md. M. R. Khan et al.	SAM-40 (Math Task)	EEG	DWT, Filter bank, Two-stage hybrid classifier	95.56	Small sample size required data augmentation; generalizability to other stressors is unproven.
8	2026	A. Singh et al.	MODMA	EEG, Audio	ResKAN encoder, Cross-attention mechanism, KAN	99	High computational resource demands for KAN networks; pending diverse clinical trials.
9	2023	Not Specified	Frontal Lobe (38	EEG	Majority Voting Ensemble (SVM, KNN, Naïve	93.85	Heavy reliance on manual feature extraction limits fully automated scaling.

			Subjects)		Bayes)		
10	2023	Not Specified	VR Horror / IQ Test	EEG	K-Means Clustering + Support Vector Machine (SVM)	Not Specified	Lack of objective ground truth labels; heavily relies on subjective evaluations.
11	2026	Nazmus Sakib et al.	IUB Students (PSS-10)	EEG	Bagged Trees Classifier	Not Specified	Generalizability is limited as participants were exclusively university students.
12	2024	Vaishali Darbar et al.	Not Specified	EEG	Deep Neural Network (DNN) & ML Classification	Not Specified	The "black-box" nature of deep neural networks lacks interpretability for clinical diagnosis.
13	Not Specified	Not Specified	Not Specified	EEG	Power Spectral Density (PSD) Asymmetry Analysis	N/A	High individual variability makes establishing standard universal baselines difficult.
14	Not Specified	Not Specified	Multiple (Review)	Nano EEG	Review of Management and Detection Techniques	N/A	Manufacturing complexities; long-term biocompatibility and durability of nano-sensors are largely unknown.
15	Not Specified	Not Specified	Not Specified	EEG	Interpretable Machine Learning Framework	Not Specified	Balancing high predictive accuracy with interpretability across diverse demographics remains a challenge.
16	Not Specified	Anagha Acharya et al.	Multiple (Review)	EEG	Review of ML clinical tools for Postpartum Depression	N/A	Scarcity of longitudinal EEG data during and post-pregnancy due to ethical data collection constraints.
17	Not Specified	Urvashi P. Shukla et al.	Not Specified	EEG	Deep Neural Network (DeepNet)	Not Specified	Multi-channel setups are cumbersome and intrusive for daily wear and real-world monitoring.
18	Not Specified	Yashada Nikam et al.	Multiple (Survey)	Physiological/ EEG	Survey of Feature Extraction & ML models	N/A	Lack of standardized pre-processing pipelines makes cross-study comparisons difficult.
19	Not Specified	Not Specified	Multiple (Review)	EEG	Review of Current Strategies and Future Aspects	N/A	Transitioning from lab-grade equipment to reliable consumer-grade wearables with high signal-to-noise ratios.
20	2020	Natalya	Siberi	Resti	Delta/Gamma	N/A	Confounding variables (cultural/lifestyle

		Milakhina et al.	an Migrants	ng State EEG	Spectral Marker Analysis		changes) not fully isolated; highly specific to polar regions.
21	2023	S. R. Chowdhury et al.	Rat Model	EEG, tDCS	Preclinical Electrode Validation	N/A	Direct translation from animal (rat) models to human clinical trials is complex; long-term effects of tDCS need tracking.